

An LLM Agent for the Selection, Validation and Interpretation of Statistical Methods

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Programme	Capstone Project
Document	Project Proposal
Date	11 September 2026

1. Problem and motivation

A large language model asked to analyse a dataset will almost always produce an answer. Whether that answer is *valid* is a different question, and one the model is poorly placed to police. A data file cannot reveal whether rows are independent, whether the same subject appears twice, whether observations are ordered in time, or whether measurements cluster within groups. These are facts about the study design, not about the numbers, and they determine which statistical method is admissible.

The consequence is a specific and under-examined failure mode. A mathematically fluent system with no view of study design will confidently fit a Poisson regression to autocorrelated hourly counts, or report a correlation across observations that are not independent. The output is well-formed, the arithmetic is correct, and the conclusion is invalid. Existing evaluations of LLMs on data analysis largely measure whether the code runs and the arithmetic is right, which does not detect this class of error at all.

2. Research question

Does an explicit statistical decision protocol improve an LLM agent's ability to select and execute valid statistical methods, compared with direct LLM advice and with tool access that has no protocol governing its use?

The contribution is not the interface, nor the number of supported tests. It is controlled evidence that design-aware routing, targeted diagnostics and verification improve validity over ordinary LLM tool use. Stating the question this way fixes what must be held constant: if the protocol arm and the tool arm do not share tools, prompts and report schema, any difference between them is uninterpretable.

Hypotheses

- **H1.** The structured agent selects an admissible method more often than a tool-enabled baseline with no protocol.
- **H2.** The structured agent abstains more reliably on designs that no supported method fits — the safety-critical case.
- **H3.** Enforcing provenance at the schema level eliminates fabricated numerical detail in the structured arm, where the unconstrained arms retain a measurable rate of it.

3. Scope

The method library is deliberately closed at fourteen methods plus abstention: three two-group tests, three k-group tests, two categorical tests, two correlation measures, and four regression models. A closed library is what makes the selection task scorable — an open-ended one would turn method choice into an essay-grading problem.

Paired and repeated-measures designs, mixed models, time series, survival analysis and zero-inflated models are explicitly out of scope. They are not omitted quietly: they enter the benchmark as cases whose correct answer is to abstain, which converts a scope limit into a measurable behaviour.

There is deliberately no general code-execution tool. It would let the model bypass the method policy entirely and would make traces incomparable across the three arms, destroying the comparison the project exists to make.

4. System design

4.1 Three systems over one shared core

Three drivers are built over a single shared core. Every driver imports the same tool registry, the same result store and the same report schema; only control flow differs. Baseline fairness is therefore provable from the repository layout rather than asserted in prose — there is no code path on which a baseline could have been disadvantaged.

System	Description
A — Direct	Receives the same step-one diagnostic output the structured agent receives, not a hand-picked summary, and answers without tools.
B — Tools	Identical tool definitions and identical report schema to System C, with no ordering imposed on their use. This is the baseline the research question turns on.
C — Protocol	A deterministic state machine: parse the problem, validate the design, enumerate candidates, run only diagnostics that could change the decision, select or abstain, execute, verify.

Table 1. The three arms of the comparison.

4.2 The provenance contract

The strongest claim the system makes is that **the model cannot write a number**. Rather than inspecting a finished report for fabrication, fabrication is made structurally impossible: every tool return is registered in a run-scoped store under a stable key; the report schema rejects raw numeric literals in prose; the model emits reference templates that are substituted at render time; and an unknown reference fails the run outright.

This must be reported honestly rather than as a headline result. For the structured arm, numerical fidelity becomes an *architectural guarantee*, not an empirical finding. Its empirical counterpart is the rate at which the model reaches for a reference that does not exist. For the two baselines nothing is enforced, so fidelity remains a genuine measurement there and the comparison stays meaningful.

5. Benchmark design

64 cases, of which 12 have no admissible in-scope method and must be declined. Cases are never hand-authored: one declarative registry entry produces each complete case package, so the benchmark regenerates from source.

For synthetic cases the gold label is derived from the *realised sample* rather than from the generator's intent. Design facts come from the generator parameters, because independence and pairing are properties of the design; distributional facts come from the data, because that is what a correct method choice must respond to. A generator asked for a normal sample that happened to draw a skewed one must be labelled for the sample it drew.

Source	Rows	Role in the benchmark
Bank Marketing (UCI 222)	45,211	Categorical association, logistic regression, skewed group comparison
Online Shoppers (UCI 468)	12,330	Association, categorical tests, logistic regression
Seoul Bike Sharing (UCI 560)	8,760	Four design-hazard cases where hourly ordering breaks independence
Student Performance (UCI 320)	649	Group comparison, ordinal traps, ordinary least squares

Table 2. Public data sources. Seoul Bike supplies the sharpest test: tables that look entirely suitable for Poisson regression, correlation or ANOVA, where the correct answer is to abstain.

6. Evaluation design

The benchmark is split into development and held-out partitions and the split is frozen before any held-out execution. Prompts, model identifier, code and tools are frozen at the same point. This is the control against the most damaging threat to the result — that apparent performance reflects prompt tuning against the test cases.

Metrics

- **Method-selection accuracy** against an accepted-method set, reported with Wilson intervals.
- **Abstention recall** on design-hazard cases, and the complementary **unsafe-selection rate**: fitting a real method to an invalid design.
- **Numerical fidelity** and the provenance rejection rate.
- **Assumption coverage** and unsupported-inference rate in the written interpretation.
- **Run-to-run variability** across repetitions, reported rather than suppressed.

The primary comparison is paired at case level. Repetitions collapse to a per-case majority before testing, because McNemar's test assumes independent pairs and two runs of one case are not two cases. Differences are reported with a case-level bootstrap interval alongside the exact test.

Determinism deserves a note. The `temperature` parameter has been removed on current models and returns an error if sent. Determinism is therefore controlled by pinning the model identifier, fixing the effort level, freezing a prompt hash recorded in every run manifest, and measuring the residual variance across repetitions rather than claiming to have eliminated it.

7. Schedule and acceptance gates

The schedule uses weekly acceptance gates. A gate is a runnable artefact or a validated dataset, not a status note.

Week	Build work	Evaluation work	Acceptance gate
1	Freeze scope; schemas, repository, case validator	Acquire four UCI sources; 16 development cases and initial labels	Raw data reproducible; 16 development cases pass validation
2	Typed diagnostics; execution dispatchers; unit-test 14 methods	Complete all 64 cases, oracles and reviewer pass; lock the split	Every method unit-tested; all cases valid; label disagreement resolved
3	Direct baseline, tool baseline, state machine, trace logging, UI	Development cases only; refine routing from documented errors	All three systems finish every development case; no untraced claim
4	Verifier, one-revision rule, report renderer, reproducibility manifest	Blinded development review; freeze prompt, model, code and runner	Development accuracy $\geq 80\%$; numeric fidelity $\geq 98\%$; candidate tagged
5	Demo and packaging; presentation defects only	Execute the held-out evaluation; score, bootstrap, error analysis	Results reproducible from raw logs; demo handles success and abstention

Table 3. Weekly acceptance gates.

8. Risks and controls

Risk	Material consequence	Control
The dataset lacks design facts	A mathematically plausible but invalid method is selected	Mandatory design card; abstain when independence, pairing, time order or clustering is unknown
Assumption checks become mechanical	Diagnostic p-values act as simplistic switches	Policy consults multiple diagnostics and makes the estimand explicit
Ambiguous answer keys	Accuracy measures annotator preference rather than validity	Accepted-method sets, written rationale, second review, recorded disagreement resolution
Benchmark leakage	Apparent performance reflects tuning on test cases	Separate gold path; stratified frozen split; prompt and code freeze before any held-out execution
The model fabricates numerical detail	Plausible but false results reach the final report	Tool-only numerical policy, result identifiers, strict verifier, schema-level provenance
Real data invites causal claims	Interpretation exceeds an observational design	Causal-language linting and an explicit randomised or observational field
Scope expands to advanced statistics	The schedule collapses	Unsupported designs are benchmarked as abstentions; extensions are documented, not implemented
Provider nondeterminism	Results are unstable or hard to reproduce	Pinned model, fixed effort, frozen prompt hash, repetitions, full run manifests

Table 4. Risk register with controls.

9. Deliverables and success criteria

- A working application accepting a dataset, a question and a design card, returning a traceable report or a reasoned abstention.
- A tool library of fourteen methods with effect sizes, intervals and diagnostics.
- A benchmark of 64 validated case packages with gold labels, oracles and a frozen split.
- Three system variants, complete run logs, scorers, result tables and an error taxonomy.
- A capstone report presenting results, error analysis and limitations.

The project succeeds if the structured arm exceeds the tool-enabled baseline by at least ten percentage points of method-selection accuracy on the held-out partition, with the difference supported by a paired test, and if every numerical claim in every generated report maps to a recorded tool output. It fails honestly — and is reported as failing — if the difference does not materialise.